CISS245: Advanced Programming
Quiz q02

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Q1. Complete the following C++ statement so that `r` has a random integer value in the interval $[5, 13]$, i.e., the value is 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12.
Answer:

```
int r = rand() % 9 + 5;
```

Q2. Complete the following C++ statement so that `r` has a random integer value in the interval $[-100, 100]$.
Answer:

```
int r = rand() % 201 - 100;
```

Q3. Complete the following C++ statement so that `r` has a random even integer value in the interval $[-100, 100]$.
Answer:

```
int r = (rand() % 101) * 2 - 100;
```

Q4. Complete the following C++ statement so that `r` has a random double value in the interval $[-1.1, 8.6]$.
Answer:

```
double r = rand() % 10000 / 10000 * (8 + 1.1) - 1.1;
```

Q5. Write a *complete program* that rolls two (random) dice until two sixes are obtained. Here's a test run

```
(1, 1) (2, 1) (6, 6)
```

Answer:

```
#include <iostream>

int main()
{
  srand((unsigned int) time(NULL));

  while(d1 < 6 && d2 < 6)
  {
    int d1 = rand() % 6 + 1, d2 = rand() % 6 + 1;
```

```
    std::cout << '(' << d1 << ``, `` << d2 << ``) ``;

    if(d1 == 6 && d2 == 6)
    {
      break;
    }
  }
  return 0;
}
```

Q6. Write a *complete program* that rolls two (random) dice until two sixes are obtained. Furthemore, if the first roll is a 1, you do not roll the second die, but you continue to the next try. Here's a test run

```
(2, 3) (5, 3) (1) (4, 5) (6, 6)
```

Notice that in the third try, the first die is a 1, so you forget about rolling the second die and go on to the fourth try.
Answer:

```
#include <iostream>

int main()
{
  #include <iostream>

int main()
{
  srand((unsigned int) time(NULL));

  int d1 = 1, d2 = 1;

  while(d1 < 6 && d2 < 6)
  {

    d1 = rand() % 6 + 1;

    if(d1 == 1)
    {
      std::cout << '(' << d1 << ``) ``;
    }
    else
    {
      d2 = rand() % 6 + 1;
      std::cout << '(' << d1 << ``, `` << d2 << ``) ``;
    }

    if(d1 == 6 && d2 == 6)
    {
```

```
      break;
    }
  }
  return 0;
}
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.